ISHRAK LATIF

+61 489 154 254 | ishrak.latif96@gmail.com | ishraklatif.github.io | github.com/ishraklatif

PROFESSIONAL SUMMARY

AI/ML Engineer with 7+ years across production ML systems, NLP pipelines, and edge AI — completing an MSc in Artificial Intelligence at Monash University (graduating 2026). Specialises in RAG system design, LLM evaluation, and ML deployment across cloud and edge, including fully offline on-device LLM and vision systems on NVIDIA Jetson hardware. Active researcher with Monash VISTA on low-power edge AI hardware for smart city and environmental monitoring applications. Combines deep technical delivery with stakeholder advisory experience at government and enterprise level.

TECHNICAL SKILLS

RAG & Retrieval: Hybrid BM25+FAISS, fastembed, ChromaDB, Azure AI Search (hybrid + RRF fusion), CrossEncoder reranking, LLM-as-Judge, Cohen's kappa

LLMs & Fine-tuning: LoRA/PEFT (T5, GPT-2), HuggingFace Transformers, llama.cpp, Groq API, Ollama, LangChain, LangGraph, prompt engineering, query rewriting

Edge AI & CV: YOLOv8 (distillation, ONNX export), OpenVINO, NNCF INT8 quantisation, ByteTrack/BoT-SORT, Depth Anything V2, MediaPipe Pose, camera calibration, point clouds, Jetson Xavier NX (JetPack 5), Jetson Orin Nano, Raspberry Pi 5, Intel NUC (CPU/iGPU inference)

ML & Data: PyTorch, scikit-learn, LightGBM, MLflow, PySpark, numpy, pandas, GRU/seq2seq, Random Forest

Deployment & Infra: FastAPI, Flask, Docker, Kubernetes, Fly.io, AWS EC2, Azure (AKS, Key Vault, AI Search, Azure ML), Supabase, PostgreSQL, SQLite, CI/CD (GitHub Actions, Jenkins)

Languages: Python, SQL, JavaScript, Bash

RESEARCH EXPERIENCE

AI Research Engineer | Monash VISTA, Monash University, Melbourne, AU *Aug 2025 – Present*

- Contributing to EdgeSight — a low-power edge AI hardware platform for running vision ML/DL models and small language models entirely on-device across 5 field pilots: smart cities, water/stormwater, bush/fire, smart transportation, and healthcare.
- Synthesized a 10-phase project execution plan (Hardware, Software, Business & Admin tracks) spanning architecture through field pilot deployment, consolidating the official roadmap and supplementary research brief.
- Evaluated edge compute platforms (NVIDIA Jetson Orin Nano 8GB, Raspberry Pi 5 + Hailo-8L, Rockchip RK3588, Intel N-series/Lunar Lake) against power envelope, NPU throughput, and PyTorch/ONNX deployment criteria for off-grid solar vs mains-powered profiles.
- Designed a model compression pipeline: teacher-student knowledge distillation (YOLOv8m/l → YOLOv8n-class student), exported via PyTorch→ONNX→OpenVINO, quantised to INT8 with NNCF, benchmarked for latency/accuracy tradeoffs across CPU and iGPU.
- Developed an on-device 3D digital-twin prototype on a Jetson Xavier NX smart camera, combining metric monocular depth, calibrated point-cloud reconstruction and real-time human pose tracking, running fully offline on the edge hardware.
- Assessed in-hand prototype hardware (Intel NUC10i7FNH, Comet Lake, no NPU) against project power constraints; designed a fallback CPU/iGPU inference strategy with OpenVINO.
- Produced hardware spec sheets, risk register, and solar/battery power budget calculations for cross-functional team alignment.

PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer | Aztech Valley Ltd, Bangladesh *Sep 2022 – Jul 2024*

- Architected and deployed a production NLP triage platform serving 200+ clinical staff, reducing average emergency response time by 20%.
- Built RAG pipelines using BioMistral and PubMedBERT with ChromaDB vector indexing, improving medical query accuracy by 25% and reducing retrieval latency by 40%.
- Engineered PySpark EHR preprocessing pipelines on Azure supporting 1.5M records/hour with distributed fault-tolerance.
- Automated CI/CD across Kubernetes clusters (Docker, Jenkins, MLflow), cutting manual deployment effort by 60%.
- Implemented GDPR-compliant data encryption using Azure Key Vault; achieved 100% audit pass rate across all security evaluations.

AI Consultant | Dhaka National Medical Institute & BD Thai Ltd, Bangladesh *Jul 2023 – Jul 2024*

- Advised hospital directors and government stakeholders on AI-based medical technology acquisitions, reducing procurement cycle time by 15%.
- Designed AI modernisation roadmaps presented to the Bangladesh Health Ministry; secured ministerial approval and funding for a national hospital digitisation initiative.

Lieutenant — Cybersecurity & Operations | Bangladesh Army, Bangladesh *Jan 2016 – Jun 2021*

- Led 120+ personnel in high-pressure operational environments; conducted penetration testing and cybersecurity assessments on MLRS fire-control networks.

PROJECTS

CAPYBARA STATION — On-Device LLM Wayfinding Assistant

- Architected a fully offline, two-device edge AI kiosk: an ADLINK NEON (Jetson Xavier NX) hosts the Flask/Gunicorn app and avatar display, while a second Jetson Xavier NX serves Llama 3.2 3B (Q4_K_M) via llama.cpp over a private direct link, with no cloud dependency.
- Designed a hallucination-safe navigation pipeline: the LLM produces JSON-schema-constrained decisions, and deterministic Python selects routes from a verified knowledge base with accessibility-aware ranking and authored response templates.

- Built a streaming chat API (Server-Sent Events) with cancellation, replay, privacy-preserving session storage and name redaction, plus a dependency-free JavaScript frontend with mood-driven avatar states, validated by 300+ automated Python and JavaScript tests.
- Designed a dual-mode interface: Navigator offers one-tap destinations, visitor-set accessibility preferences and instant alias matching (LLM as fallback); Friend provides persona-driven conversation in a single model call.

Python, Flask, llama.cpp, Llama 3.2 3B, Jetson Xavier NX, ADLINK NEON, SSE, SQLite

NEON DIGITAL TWIN — Edge 3D Scene & Human Pose

- Built a camera-to-3D-scene prototype on an ADLINK NEON smart camera (Jetson Xavier NX, JetPack 5.0.2) that turns a single RGB stream into a coloured metric point cloud and tracks a 33-joint human skeleton, all on-device.
- Implemented the full pipeline: GStreamer capture/replay, checkerboard intrinsic calibration (0.57 px RMS), Depth Anything V2 metric-depth inference on CUDA, and back-projection to PLY point clouds (143K+ points per frame).
- Integrated MediaPipe 33-joint pose and person masks on the device CPU (~0.26 s/frame) with timestamp-aware smoothing, keeping the static scene and moving people separate, with explicit handling when a person is lost and re-acquired.
- Built a live HDMI dashboard (RGB/pose, depth, point cloud, rotatable 3D skeleton) with a single camera owner and depth inference in a separate, timeout-guarded process.
- Set up isolated ARM64 runtimes from pinned, checksum-verified offline wheel sets and models, leaving the vendor CUDA/OpenCV stack untouched and requiring no internet at runtime.
- Wrote 58 synthetic and regression tests covering projection, crop/resize mapping, mask alignment and state transitions; every output records its frame/time, coordinate system and calibration.

Python, PyTorch, Depth Anything V2, MediaPipe, OpenCV/GStreamer, NumPy, Jetson Xavier NX

CAPY RAG — AI Education Chatbot

- Built a RAG-powered AI-education chatbot (capybara persona) for high school students (ages 14–18) covering 8 AI/ML topics including neural networks, LLMs, ethics, and careers.
- Implemented hybrid BM25+dense retrieval with CrossEncoder reranking; structured JSON responses (answer, emotion, fun_fact) to support a future video/audio avatar phase.
- Notebook version (Groq API): query rewriting, few-shot prompted generation, LLM-as-Judge evaluation with self + cross-model judging and Cohen's kappa vs human scores.
- Persona tuned for teen relatability: self-aware capybara humor, sparse current slang (capped at 1–2 per response), platform-relevant analogies (For You page, Discord, gaming).

Python, BM25, FAISS, Ollama, Groq, LLM-as-Judge

PORTFOLIO CHATBOT — RAG Backend

- Built and deployed a production RAG chatbot backend on Fly.io serving ishraklatif.github.io — retrieves context from resume + blog posts (Supabase) using BAAI/bge-small-en-v1.5 embeddings.
- Blog system with admin CRUD panel; /reload endpoint refreshes the RAG index without redeployment; LLaMA-3.1-8B via HuggingFace router for generation.

FastAPI, fastembed, Supabase, Fly.io, HuggingFace

EDGESIGHT EDGE AI PLATFORM

- Designed knowledge distillation pipeline (YOLOv8m→YOLOv8n), ONNX→OpenVINO export, and INT8 quantisation with NNCF for CPU/iGPU deployment on power-constrained hardware.
- Evaluated hardware platforms for off-grid solar deployment; produced power budget calculations and technical documentation for a 5-pilot field deployment roadmap.

YOLOv8, ONNX, OpenVINO, NNCF, Jetson Orin Nano

ASX EQUITY RETURN FORECASTING

- Cross-sectional return forecasting across ASX 50 with a 260-feature panel (245K+ samples, 2005–2026). Test IC = +0.054, ICIR = +0.314, annualised L/S Sharpe = +0.429.
- Resolved T4 GPU OOM in TFT's Variable Selection Network by redesigning to a VSN-free linear encoder, reducing GPU memory ~6× while preserving the temporal attention stack.

Python, LightGBM, PyTorch

NLP QUESTION CLASSIFIER

- Built full-stack NLP classifier (6 TREC categories) comparing custom Transformer encoder vs BERT prefix tuning. Prefix tuning achieved 94%+ accuracy training <1% of BERT parameters.

Python, PyTorch, HuggingFace, FastAPI, React

EDUCATION

Master of Science in Artificial Intelligence | Monash University, Melbourne, AU *2024 – 2026 (Expected)*

- Key coursework: NLP, Computer Vision, Multi-Agent Systems, Planning & Automated Reasoning, Applied ML.
- Capstone: Calmer Commute — sensory-aware navigation app for Melbourne CBD; AI/ML owner (crowd forecasting, route scoring, AWS EC2 deployment).

Bachelor of Science in Computer Science & Engineering | Bangladesh University of Professionals, Dhaka, BD *Jan 2022*

INDUSTRY EXPERIENCE

Developer Lead & AI Implementation Lead | Monash University Industry Experience

- Leading development and AI implementation for the university Industry Experience project.
- Won first place at the Industry Experience Expo. View project ePortfolio